

Integration of Classy-Blocks and OpenFOAM within Blender

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Weekly Progress Report

OpenFOAM GUI

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1 Introduction

2 Research Gap

3 Problem Statement

4 Work Done

Introduction

- ◀ **OpenFOAM:** A C++ toolbox for CFD using finite volume methods.
- ◀ **Classy-Blocks:** A Python library for procedural `blockMeshDict` generation, mimicking modern 3D CAD modeling.
- ◀ **Blender Integration:** Utilizing Blender as a frontend for geometry extraction and mesh visualization using `pyvista`.

Research Gap

- ◀ **Gap 1:** Lack of native Blender 5.x support for advanced Python meshing libraries like `classy_blocks`.
- ◀ **Gap 2:** Absence of an integrated "One-Click" solution for geometry extraction, mesh building, and quality reporting within a unified environment.

Problem Statement

- ◀ To develop a Blender Add-on that leverages `classy_blocks` and `pyvista` to automate OpenFOAM case setup, allowing users to extract geometry, generate high-quality meshes, and execute solvers without leaving the Blender interface.

Work Done: Sunday (08-03-2026)

- ▶ **Code Audit:** Reviewed all core prototype files including `operators.py`, `mesh_builder.py`, and `foam_runner.py`.
- ▶ **Testing Phase:** Executed full test suite (`test_classy.py`, `test_subprocess_foam.py`, etc.) and identified pass/fail metrics.
- ▶ **Environment Confirmation:** Verified Blender 5.x Python environment successfully identifies `classy_blocks`, `pyvista`, and `numpy`.

Work Done: Monday (09-03-2026)

- ◀ **Documentation:** Created DEV_NOTES.md mapping current file architecture and distinguishing stubbed functions from live code.
- ◀ **Infrastructure:** Initialized geometry_extractor and mesh_builder modules.
- ◀ **Import Management:** Integrated stubs into operators.py to prepare for operator logic wiring.

Work Done: Tuesday (10-03-2026)

- ▶ **Operator Wiring:** Implemented `case_path` retrieval from `context.scene.classy_mesh_props`.
- ▶ **Path Logic:** Computed dynamic `output_path` for `system/blockMeshDict` based on the user-selected case path.
- ▶ **Execution Chain:** Linked `extract_geometry` (returning hardcoded specs) to the mesh builder pipeline.

Work Done: Wednesday (11-03-2026)

- ◀ **Error Handling:** Added try/except blocks to the operator chain to capture and report runtime errors.
- ◀ **UI Reporting:** Enabled success/error feedback via the Blender `self.report()` system.
- ◀ **Validation:** Confirmed that valid case paths trigger the full backend chain, ensuring robust communication between Blender and Python modules.

Work Done: Thursday (12-03-2026)

- ▶ **Foam Runner Integration:** Integrated `run_blockmesh` to execute OpenFOAM commands via subprocess.
- ▶ **Mesh Quality Analysis:** Implemented logic to parse `stdout/stderr` for non-orthogonality and skewness.
- ▶ **Safety Guard:** Added path validation to prevent execution errors when `case_path` or `bashrc_path` is missing.